

Foundations of Artificial Intelligence - 2026.1

Atomic Decisions under Uncertainty

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Decision Making Situations

Decision Situation Matrix

		Action Effects		
Horizon		Certainty	Risk	Uncertainty
	Atomic	✓	✓	NOW
	Sequential	✓	✓	

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Motivation

Consider this problem: Every day you can choose between cafeteria 1 and 2, each of which offers only one menu. The menus are different every day and meet your taste with a score between 1 and 5. Once you went to one cafeteria, you cannot change your mind.

The Bernoulli principle tells you that ...

... you should pick the cafeteria with the highest expected utility (average rating).

The problem is that you do not know in advance which meal is served by which cafeteria that day, and since you are new to the place, you also don't know with which probability they are served.

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Motivation

This is a decision situation under **uncertainty**, because we neither know which meal we will receive nor (at least in the beginning) how much we like them.

On the first day, we cannot make any rational justified choice.

But over time we will **learn** in which cafeteria we receive better meals, on average.

Effectively, what we do is to convert a decision of uncertainty into one under **risk**.

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Problem Definition

Consider an infinite *episodic* game in which we can choose, in each round $t = 1, 2, \dots$, an action $A_t \in \{1, \dots, K\}$, each of which associated with an unknown **reward distribution** $\mathbb{P}_i$.

Choosing action A_t at time t samples a reward $R_t \sim \mathbb{P}_{A_t}$.

The expected reward for each action a (independent of the time step) is:

$$q(a) := \mathbb{E}[R_t | A_t = a] = \mathbb{E}[\mathbb{P}_a]$$

The Bernoulli Principle tells us that we should always pick an $a^* \in \arg \max_a q(a)$.

Challenge: Since we do not know $\mathbb{P}_a$, we do not know $q(a)$.

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Problem Definition - Offline Scenario

If we knew $\mathbb{P}_a$ for all arms, we could compute or at least approximate $q(a)$.

Then picking the optimal action would be easy.

We could try to learn $\mathbb{P}_a$, but since this is only used for $q(a)$, we can learn $q(a)$ directly.

So we could give some **finite time horizon** T to learn the true values $q(a)$ and then always use the arm we think is best.

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Estimating Average Reward

How to estimate $q(a)$ based on choice A_t and observations R_t up to time t ?

An intuitive way is to take the mean of rewards observed for arm a up to time t :

$$\hat{q}_t(a) := \frac{\sum_{s=1}^{t-1} R_s \cdot 1_{A_s=a}}{\sum_{s=1}^{t-1} 1_{A_s=a}}$$

The estimate $\hat{q}_t(a)$ itself is a *random variable* with nice properties:

1. it is **unbiased**, i.e. $\mathbb{E}[\hat{q}_t(a)] = q(a)$
2. it is **consistent**, i.e. $\lim_{t \rightarrow \infty} \hat{q}_t(a) = q(a)$ a.s.
3. it quickly concentrates around $q(a)$: $\mathbb{V}[\hat{q}_t(a)] = \mathbb{V}[\mathbb{P}_a]/t$

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Estimating Average Reward

A nice property of the empirical mean is that it can be updated incrementally:

$$\hat{q}_n(a) := \hat{q}_{n-1}(a) + \frac{1}{n}[R_n - \hat{q}_{n-1}(a)]$$

where we simplified notation a bit to say that n is the time n -th time step in which action a was chosen.

This way, the time to compute $\hat{q}_n(a)$ is constant and does not grow with n .

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Exploration vs. Exploitation

Since we do not know a^* in the beginning, we must somehow first **explore** the options to make sure that the $\hat{q}_t(a)$ are close to $q(a)$.

As soon as $\hat{q}_t(a) \approx q(a)$, we identify a^* and we can **exploit**, i.e. only draw a^* .

But how do we know whether $\hat{q}_t(a) \approx q(a)$?

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Balancing Exploitation and Exploration

It is *impossible* to *ever* be sure that one has identified the best arm.

Why?

That means: Exploration and Exploitation must be interleaved:

1. at each point of time, there must be a strictly positive probability of exploring, but
2. depending on the experience, higher probability should be given to the arm that is *supposedly* optimal.

Intuitively: Pull $\arg \max_a \hat{q}_t(a)$ but maintain some mechanism for exploration.

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Problem Definition

Any strategy we may use creates a (usually stochastic) sequence $(A_t)_{t \in \mathbb{N}}$.

How do we measure the quality of such a strategy?

Some possible ways of scoring a finite sequence $A_1, \dots, A_t$ are:

1. the **average reward**: $\frac{1}{t} \sum_{s=1}^t R_s$
2. the **rate of optimal choice**: $\frac{1}{t} \sum_{s=1}^t 1_{A_s=a^*}$
3. the **cumulative regret**: $\sum_{s=1}^t q(a^*) - q(A_s)$

The latter two can only be computed if we (the experimenters) actually know the $\mathbb{P}_a$.

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The Greedy Strategy

The simplest strategy is to try each arm once and then always pull $\arg \max_a \hat{q}_t(a)$.

The Greedy Strategy is exploitation-only, virtually there is no exploration beyond initialization.

This is ideal if $a^* = \arg \max_a \hat{q}_t(a)$ but can be pretty bad otherwise.

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The ε -greedy Strategy

A simple modification to consider both aspects is to define some exploration probability ε_t as a function of time t (but independent of the history).

At time t , one pulls the arm

$$\arg \max_a \hat{q}_t(a)$$

with probability $1 - \varepsilon_t$ and with probability ε_t a random arm.

A simple choice for ε_t could be

- a constant such as 0.1 or 0.01 or
- a variable that changes over time, i.e. $1/t$

Note that $\varepsilon_t = 0$ is the **greedy** strategy.

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The UCB1 Policy

A simple but effective bandit strategy is **UCB1**. Here, the confidence bound is created using a factor that balances the number of pulls of an arm in terms of the overall time elapsed so far:

$$A_t^{UCB} := \arg \max_{a \in \{1 \dots K\}} \left\{ \hat{q}_t(a) + c \sqrt{\frac{\ln(t)}{N_t(a)}} \right\},$$

where

- c is an exploration hyperparameter, often set to $\sqrt{2}$ if we know that $R_t \in [0, 1]$,
- $N_t(a)$ counts how often a has been tried at time t .

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The UCB1-Policy

The second term $c\sqrt{\frac{\ln t}{N_t(a)}}$ is often called **exploration term**.

Properties of the exploration term are:

- proportional to the time t and inversely proportional to the sample size $N_t(a)$.
- Expands slightly over time and contracts strongly for new observations.
 $\Rightarrow$ For big values we have little confidence in the estimate and inverse.
- We can hence understand this as an exploration term since it controls under which conditions we should pull an arm that appears sub-optimal.

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Asymptotic Optimality

Since every (non-oracle) policy must explore, it is not possible that the contributions to the regret drop to 0 permanently:

Even if we already identified the best arm, we still must explore every once in a while, and this will augment the regret.

It is not possible to have a regret that grows less than logarithmically in the limit.

Most strategies (including UCB1) satisfy this condition, and every strategy that keeps exploring eventually identifies the best option. However, the question is how they behave in the *short* run.

The Influence of Initial Values

How do we initialize $\hat{q}(a)$?

Possible options:

- 0
- random
- some very optimistic, maybe impossible value
- some very pessimistic, maybe impossible value

What are the effect of all of these on exploration (with UCB1)?

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Summary

- The goal for decisions under uncertainty is to turn the situation in one under risk.
- However, this is rather implicit since we simply try to directly learn the expected utilities (rather than the probabilities to then compute the expected utilities)
- The problem is that we never know when we have approximated the true expected utilities good enough.

Therefore we must interleave and trade off exploitation (use of knowledge) with exploration (complement knowledge)

- all strategies that do this must have a logarithmic regret, but some are better than others, i.e., UCB1 better than greedy- ϵ